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Pet Tracking System Using GPS with Android-Based Geofencing Method

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Abstract—The increasing interest in owning pets during the pandemic was driven by the design of a GPS tracking system with an android-based geofencing method. This system was developed to make finding these animals easier for pet owners. The system developed aims to prove that geofencing technology can make it easier to find pets with the help of technology, in addition to using an android-based application to find out the location of pets and the accuracy of the pet's distance from the home base. The methods used in developing this system are the creation of hardware using the ESP32 microcontroller and NEO7M GPS and the creation an Android-based application to make it easier for users to track the whereabouts of pets. The application also lets one know the animal's distance from the home base to the pet's location. From the tests, the average accuracy of devices with GPS-based applications is 92%; this tracking system also has disadvantages when tracking inside the building, with a delay time of approximately >3 minutes. It takes 3-5 seconds to get latitude and longitude data in open area conditions. The developed system can show precise locations but must be developed better regarding battery selection and signal problems when inside the building.

Keywords—geofencing, GPS, pet tracker, android application, GPS NEO 7M, microcontroller ESP32

I. INTRODUCTION

In recent years, the increasing number of pet owners has led to a growing concern about the safety and well-being of our beloved animal companions [1]. The unpredictable nature of pets and rising urbanization have intensified the need for innovative solutions to track and safeguard their movements effectively [2]. Leveraging the ubiquity of Global Positioning System (GPS) technology correlated to security systems and the versatility of any-based platforms, this technology is unique and creative for any research topic [3], [4]. So, we present a novel Pet Tracking System empowered by a Geofencing Method because tracking techniques have been implemented for various purposes, especially on mobile devices [5], [6].

Traditional pet tracking systems rely on manual monitoring or limited-range radio frequency identification (RFID) tags [7]–[11]. Such methods have inherent limitations,

including restricted coverage and inadequate real-time tracking capabilities [8]. Besides, some related studies adopted the tracking technology, such as using custom markers attached to the collar named BeRSTID [12], but it is limited only to monitoring eating and drinking behaviors in cats. A study by [13] also adopted the tracking technology using RFID for social behaviors. On the other hand, our proposed pet tracking system capitalizes on the widespread accessibility of GPS and the popularity of Android smartphones, offering an all-encompassing solution to pet owners for location monitoring, activity analysis, and enhanced security.

This research aims to address the challenges faced by pet owners by devising an innovative system that is not only technologically advanced but also user-friendly. The proposed system leverages the GPS module present in modern Android smartphones, enabling pet owners to track their animals with unprecedented precision and convenience. Moreover, by employing a Geofencing Method, we extend the system's functionalities to set virtual boundaries (limits) or "geofences" around designated areas, alerting the owner whenever their pet enters or exits these zones.

II. METHOD

A. Analysis

A literature study is conducted at this stage on making hardware and software prototypes, supporting theories about GPS, geofencing, modules to be used, kodular, Android, databases, and field studies, such as device assignment and modules. The output in this first stage needs an analysis of the developed system's design specifications, as seen in Figure 1.

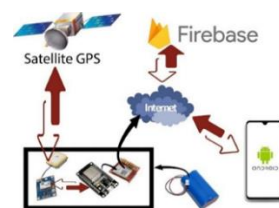


Fig. 1. System illustration for this research.

Based on Figure 1, the microcontroller (using a NEO7M GPS module) would receive the position (latitude and longitude) from the GPS satellite [14]. It would then be processed to the ESP32 microcontroller, and the 800L GSM SIM Module would make an internet connection and send it to the database – Firebase would then be sent to the Android app via an internet connection [15].

B. Design

At this stage, the design of sub-systems would be done, including the design of each module (battery module, 800L SIM module, GPS module, and modules in the application) and the design of the system, either hardware or software. The output from this stage is a system design model for designing a pet tracking system using GPS with an Android-based geofencing method.

Besides, the tracking device made in this study would use the main components, including a microcontroller as the main module, a GPS module (sensor), and a SIM module (for texting) [16]. This study used the following modules: the ESP32 microcontroller, NEO7M GPS module, GSM SIM 800L, and Battery 18650. Figure 2 illustrates the design of the developed hardware circuit.

Furthermore, in the case of software design, this research used Arduino IDE software. Arduino IDE is a default software from Arduino devices where the programming uses C / C++ language. Arduino IDE, which operates on the computer, produces a file formatted "*.ino" that would be uploaded on the Arduino board [17]. The software injected into the ESP32 microcontroller would control the performance of the hardware and connect with Firebase and Android apps. The Android application would use a kodular application with a Java-based programming language. The flowchart of software design can be seen in Figure 3.

Based on the flowchart in Figure 3, it can be explained that the application would check the limit. If the limit has not been filled, then it must be filled. If the limit is filled = 0, it must be input until the limit is not equal to 0. Next, the hardware would send the coordinate position to the database, and then the data would be processed whether the position is within the specified geofencing area. After the distance calculation process at the base, the coordinates sent earlier are still in the area, so there would only be updated coordinate information and distance from the base. However, if the coordinates are outside the geofencing area, a notification (early warning alert) would appear that pets are outside the area. Meanwhile, the physical design result can be seen in Figure 4 in the 3rd section (result and discussion).

C. Development

This stage includes making sub-systems and making systems, including coding. This stage is a continuation of the design stage. The output is in the form of making all components and starting the integration process between hardware and software that has been created.

D. Testing

Testing of process delays when outdoors and inside buildings, distance and location measurements (latitude or longitude), and responses from Android in the form of notifications about whether it can exit or not. Testing is

recorded, data processing is carried out, and conclusions and writing reports on trials are carried out.

Data sampling is the next step after creating the hardware and software. This data collection would be carried out by conducting a series of trials:

- Define the central home base and define its geofencing boundaries.
- Placing its GPS sensor device in a location that has a range of up to 100 meters from the home base ensures that the sensor can be detected, and the application can record the distance and coordinates of the sensor.
- Test the Android app to see if it can notify you if your pet passes geofencing.
- The amount of data to be retrieved is 8.

Testing criteria are carried out by researchers to state that the system that has been made is declared successful or failed, following the test criteria in the research design of a pet tracking system using GPS with an Android-based geofencing method. The data to be taken for data analysis are:

1. Getting latitude/longitude from tracking sensors.
2. Calculate the distance between the device and the home base that has been set.
3. See if the distance exceeds the predetermined geofencing and whether the app would issue a notification (distance range up to 100 meters) in testing the notification function to ensure that the developed application can issue alerts (notifications) under the specified flowchart.
4. Compare developed applications with GPS-based applications (Spotlens).
5. Conduct delay testing in the system when testing is carried out in open spaces and inside buildings. A delay here is intended to see the response of the application and the system if the test is carried out outdoors and inside buildings.
6. Testing on a system developed by obtaining latitude and longitude positions in vertical conditions, where the device would be placed at the same point at different height positions (in buildings, as well as 1st and 2nd floor).

III. RESULT AND DISCUSSION

The system developed consists of hardware (Figure 4) and software in Android-based mobile applications (Figure 5). The tests for both devices are functional tests to ensure that the two devices can be well integrated and work according to the expected goals. Here's a look at the software and hardware that has been created.

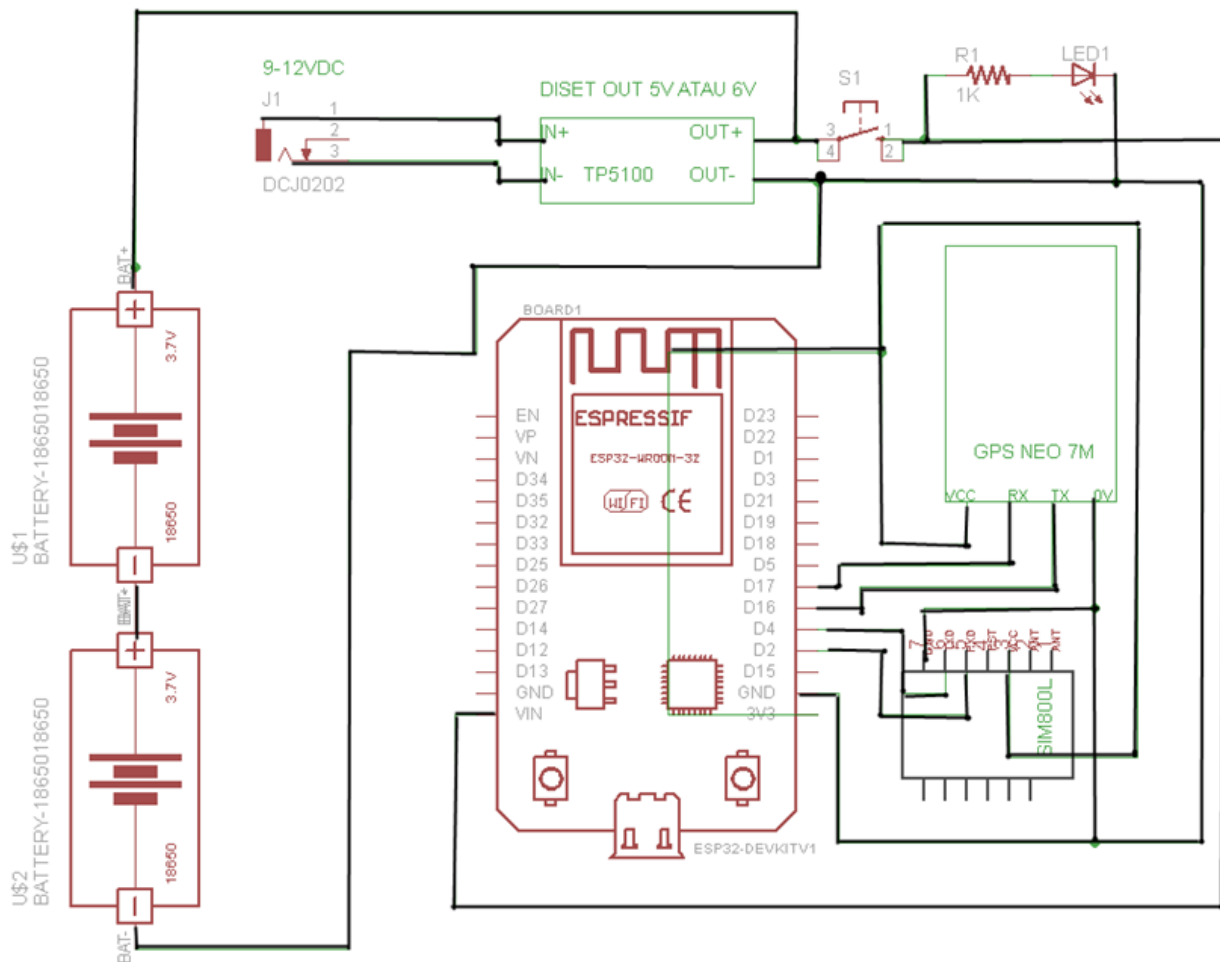


Fig. 2. Hardware schematic diagram of this study.

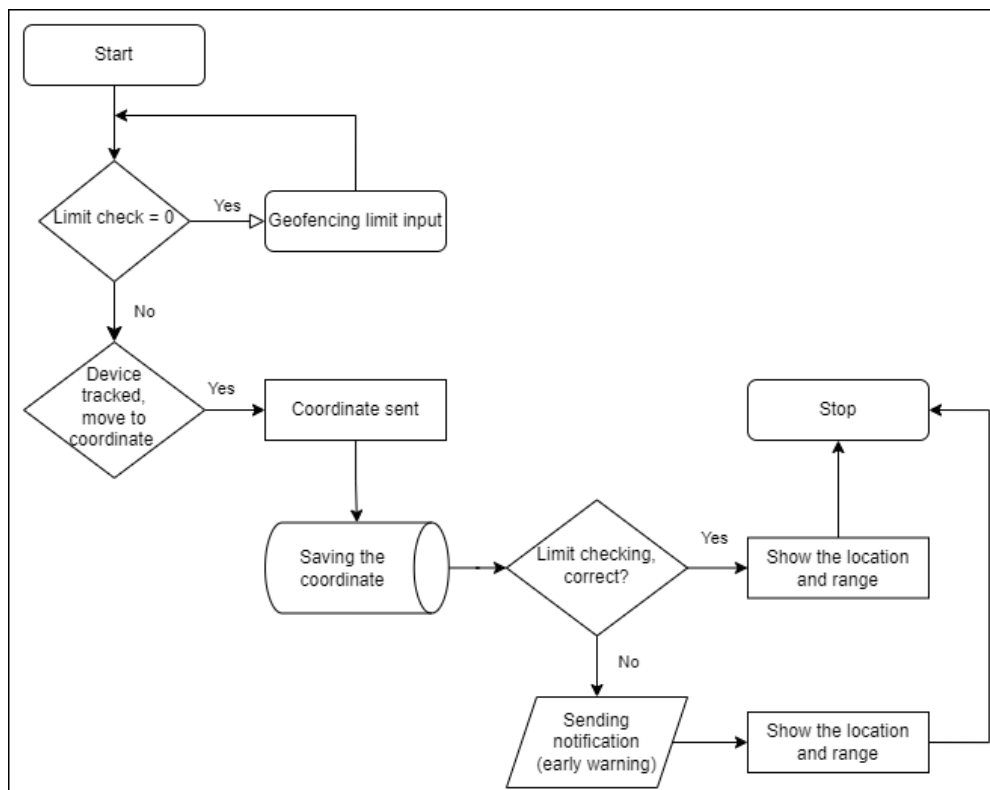


Fig. 3. Software flowchart diagram of this study.

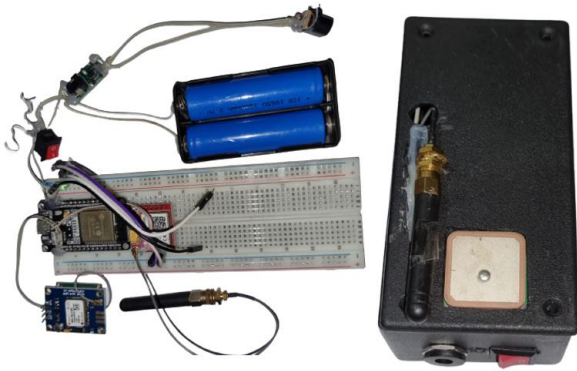


Fig. 4. The physical hardware is on the breadboard (left), and the hardware system prototype is put in the box (right).

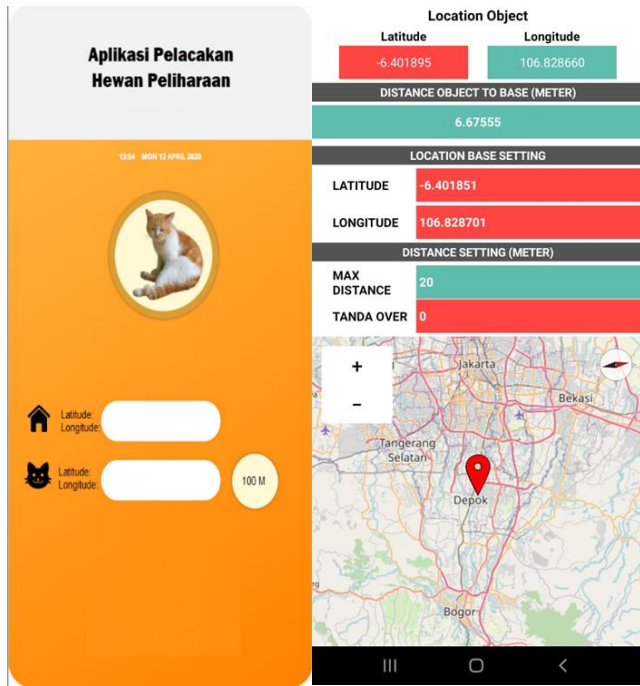


Fig. 5. The user interface (UI) on the splash screen in Indonesian (left); Main menu UI (right).

Figure 5 can be seen in the main menu (right side) to display the pet's position, which can be zoomed in/out via the +/- sign on the map. Other information on the page is the latitude or longitude position of the animal as well as the home base position (this position can be input on the settings page). Furthermore, other information is distance, where the geofencing distance can be changed as needed. The over sign on the page indicates the notification status, which would be 0 if the animal's position is less than the max distance and 1 if the animal's position is more than the max distance. The notification has a notification (beep) and the sound "your pet has gone out of area limit."

Meanwhile, the system prototype must be tested for an accurate tracking system. The first test to be performed is to ensure that the device gets the latitude/longitude position from the tracking sensor, Figure 6 being the test result, that tested from the following coordinates: -6.401851 lat. and 106.828701 long. Besides, in the second test, calculate the distance between the device and the home base that has been set, which can be seen in Table 1.

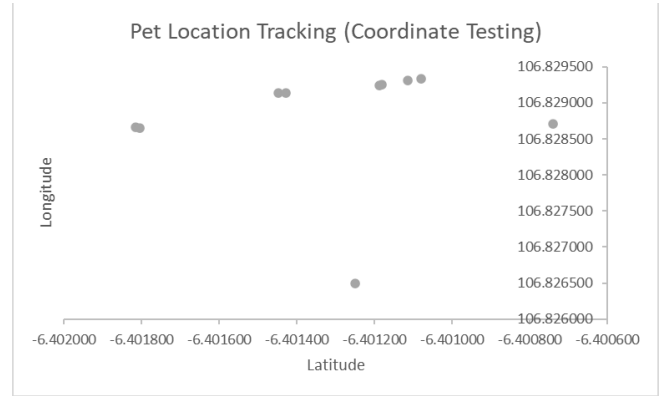


Fig. 6. Coordinate testing results.

TABLE I. RANGE CALCULATION BASED ON THE SYSTEM.

Trials	Coordinate		
	Latitude	Longitude	Range (meters) ^a
1 st	-6.401816	106.828657	6.23482
2 nd	-6.401806	106.828645	7.96694
3 rd	-6.401428	106.829139	67.56534
4 th	-6.401447	106.829136	65.86584
5 th	-6.401182	106.829251	96.16745

^a. The range is calculated from the home base.

Furthermore, the third test compares the location's position on GPS on the Spotlens app with this research. The Spotlens app was chosen as a comparison because this app is GPS-based and is commonly used in projects that use GPS or spatial-based data. The Spotlens app records the accuracy of the GPS listed between 70-100 meters from the actual point (accurate coordinate); this is influenced by the satellites used by the Spotlens app. Besides, a comparison of location position in the system with a GPS-based app (Spotlens app) can be seen in Figure 7 for the error rate and Figure 8 for the accuracy rate.



Fig. 7. Error rate testing result (in percentage of 100%).

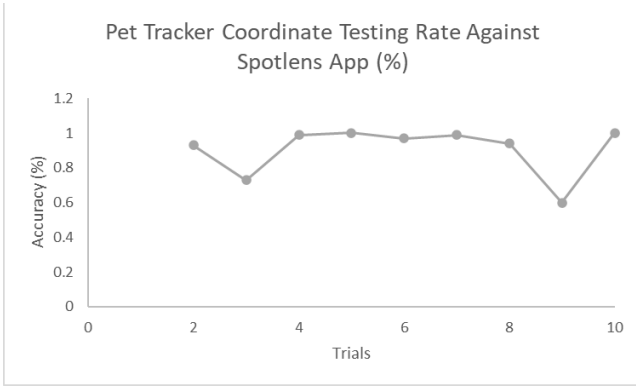


Fig. 8. Accuracy rate testing result (in percentage of 100%).

The results of the comparison test between distances on the Pet Tracker application, which is a system under development compared to the Spotlens app, can be seen in Figure 9.

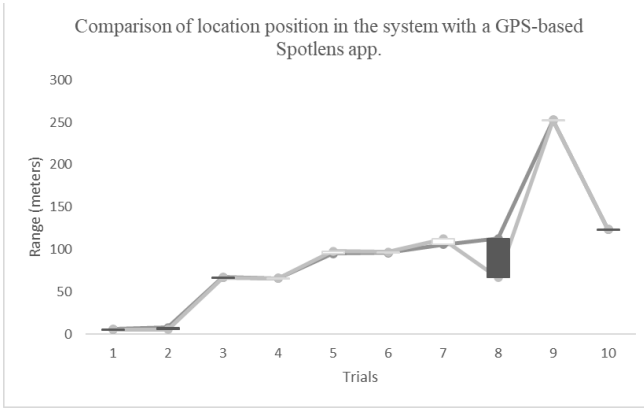


Fig. 9. Error and accuracy rate comparison result.

The results of the tests that have been carried out can be known from 10 data tests; only two data have an error percentage of more than 10%. In addition, it can also be known that the pet tracker app which is currently developed in this study, with the geofencing method, has higher accuracy because the latitude/longitude produced up to 5 digits behind the comma; this is because the longer the coordinate digits detected in spatial data show the detail of the points obtained. The other 8 data have error rates below 10%, meaning pet tracker applications and pet tracking systems use GPS with Android-based geofencing methods developed reliably.

From the comparison test between the pet tracker application and the Spotlens app, the latitude, position, and longitude of the Spotlens app are listed in the left corner of each photo taken (Figure 5). The Spotlens app provides a 5-digit number behind the comma, while the pet tracker application provides data up to 6 digits of the number behind the comma; this means that the level of detail of the data generated by the developed system is high. Besides, testing is carried out both indoors and outdoors.

The fourth test is a test of the notification function on the system, whether to issue alerts (notifications) following the specified flowchart. The notification feature in this application is considered necessary for pet owners so that owners can know if their pets are outside the predetermined limits, so it is essential to do trials, as seen in Table 2.

TABLE II. ALERT TESTING RESULT ON THE SYSTEM.

Trials	Coordinate		
	Geofencing Range (meters)	Longitude	Range (meters)
1 st	20	252.37463	1
2 nd	20	123.68194	1
3 rd	40	5.75531	0
4 th	40	17.02694	0
5 th	60	43.64435	0
6 th	60	58.12081	0
7 th	80	106.00309	1
8 th	80	96.16745	1

Based on Table 2 regarding notification testing on the system, it can be informed that while the notification status can be 0 (no) and 1 (yes) when it is 0, it means that the application does not issue notifications because the distance from the device and the base location is below the set geofencing distance. The application would issue alerts in the form of sounds (sounds) when the value is 1, which means the device is outside the set distance.

Table 2 shows that this system software that has been tested can function properly; the application evidence can show the hardware's coordinate position (latitude/longitude) location, and the application can issue alerts (notifications) according to the distance settings made.

Base location in the system app can be changed to make it easier for users (user friendly) if they move locations, for example, taking pets for a walk or even moving house. In addition, what can be changed is another distance setting for setting notifications from the app. The next test is testing the hardware and response time of the application's acceptance of latitude and longitude positions in two conditions: outdoors and in buildings (Table 3).

TABLE III. DELAY TEST RESULT ON THE SYSTEM.

Range from Home Base (meters)	Coordinate	
	Response Time in Outdoor (seconds)	Response Time in the buildings (seconds)
20	5	> 60
40	6	> 60
60	10	> 60
80	6	> 60

The response to the system would be faster when testing outdoors, while if inside the building, it would take more than 60 seconds because the device cannot directly receive the signal and cannot be processed by GPS in updating location points. The last test is to determine the position of latitude and longitude at the same point but with different heights (vertical), for example, at the same point on the 1st floor and 2nd floor; this is to find out whether latitude and longitude positions would be affected if there is a change in position (altitude). Figure 10 shows the positioning test result from different floors, tested on the 1st and 2nd floors in three different buildings within the Boolean logic.

According to Fathurrahman et al. [18], if the delay time is less than one second, the level of accuracy is better compared to GPS location and the devices. Besides, a good tracking result depends on the algorithm used, especially if using machine learning or deep learning algorithms [19], as well as image-based algorithms [20] and another relevant methods [21]. The study only used the GPS sensor and geofencing method as a conventional (non-algorithm) detection or tracking system. However, the study would be a good tool for animal monitoring and has already been implemented for livestock animals with the same method [22].

Based on the system testing, the main limitations of the system include tracking accuracy in the outdoor area and battery life. If the pet's range exceeds 100 meters, it can't be detected based on the system testing (un-coverage areas). Besides, the system would have different tracking accuracies depending on the building structure, floors, and potential metal-based materials. In addition, the estimated battery life using Battery Life Formula (1) is about 1 hour and 51 minutes. It is based on the approximate amperes output by the system, which is about 10 amperes.

$$\text{Battery life} = \frac{\text{battery capacity (mAh)}}{\text{load current (mA)}} \quad (1)$$

Furthermore, this system would be a potential challenge in the real world if it meets some requirements, such as being registered as an innovative technology in animal welfare (ethical clearances). Besides, further study is needed to decrease the hardware and battery size replacement but not limit the system's performance. In addition, it is essential to increase the tracking accuracy by combining AI-based methods, such as machine learning, deep learning, and other intelligent system methods.

IV. CONCLUSION AND FUTURE RECOMMENDATIONS

A. Conclusion

From the results of testing the pet tracking system using GPS with the android-based geofencing method that has been carried out in the previous chapter, it can be concluded that:

1. The developed system can find the location of pets (from the location of sensors shown on the map) to facilitate the search for pets with the help of technology. This demonstrable location can be detected up to less than 10 meters (tests 1 and 2 showed at 6.2 meters and 7.9 meters); this means that if paired with a pet, it can find out the position in more detail and accurately.
2. The hardware and Android app can be well connected to output the notification system according to the specified distance setting.
3. System software can adjust the location and distance according to the user's needs so that the user can customize it.
4. The device system and the developed application have an accuracy rate between 77% and 100% compared to GPS-based applications (Spotlens app), which can be interpreted as showing the location accurately and accordingly.
5. Regarding the system's response, it turns out that it would be faster to detect the latitude and longitude of the device when testing in open space, while if it is in the building, it would take > 60 seconds; this is because the signal cannot be directly received by the device and subsequently cannot be processed by GPS in updating location points.
6. The developed system cannot detect vertical position well because the system only sees the latitude and longitude of the 1st and 2nd floor at the same point.

B. Future Recommendations

From the results of testing pet tracking systems using GPS with Android-based geofencing methods that have been carried out, here are some things that can be developed for improvements from research that can be done in the future, namely:

1. The use of 18650 batteries is too large and heavy in dimensions to be applied to animals.
2. Creating a battery charging indicator is necessary, making charging more accessible.
3. Further research related to the battery to be used so that it has a longer life and faster charging.



Fig. 10. Notification test result: (A) and (B) with the status result "No" (0); (C) and (D) with the status result "Yes" (1).

4. Improved UI design (User Interface) makes it more attractive to users.
5. Further research is needed to develop a system that can perform tracking in the building.

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